THE CIRCULAR ECONOMY OF E-WASTE

KABADIWALA

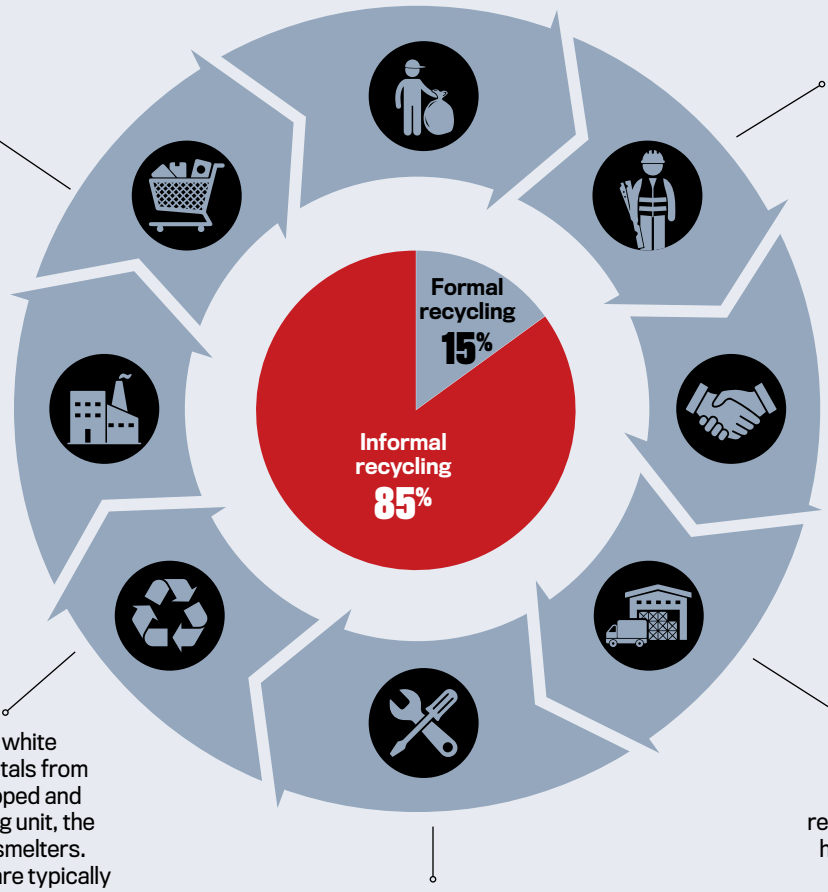
Often the starting point in the recycling chain, the local *kabadiwala* collects e-waste directly from homes

USERS

Ideally, individuals/ households should discard old devices at specific collection centres. Bulk consumers, like offices and companies, account for almost 70% of e-waste generated in the country

AGGREGATOR

Deals in larger volumes. In turn, he supplies to producer responsibility organisations (PROs) and recyclers



PRODUCER RESPONSIBILITY ORGANISATIONS

Large generators of e-waste, like companies, work with aggregators or have collection tie-ups with PROs

INDUSTRIES

The recycled materials go to various industries as raw material for manufacturing various goods

COLLECTION CENTRES

The PROs set up collection centres and direct the e-waste to recyclers. Manufacturers have to route a specified quantity of end-of-life products into recycling every year based on their annual sales. It is part of their extended producer responsibility or EPR

RECYCLERS

Steel and plastics from white goods and precious metals from mobile phones are stripped and sorted. From a recycling unit, the metals and glass go to smelters. Printed circuit boards are typically sorted, shredded and exported to specialised extractors. The non-recyclable hazardous waste is sent to treatment, storage and disposal facilities or TSDFs

RE-USE

Discarded electronics that can be re-used are refurbished and go back into the consumer market

(television sets, cameras and recorders). Half of the e-waste these products generate is iron and steel, 21 per cent plastic and 13 per cent copper, aluminium and precious metals. In addition, there are hazardous substances like mercury, lithium, lead and cadmium which, if not handled carefully, can be toxic or carcinogenic (see *Decoding E-waste*). India is now the world's third-largest source of e-waste, generating around 3.2 million tonnes annually (2019), behind only China (10 million tonnes) and the US (6.9 million tonnes), according to the Global E-waste Monitor 2020.

The situation is only likely to worsen, with the boom in the consumption of such goods threatening to add to the tonnes of waste India already generates. We sell over 150 million smartphones a year, 17.5 million television sets, 20 million pieces of audio equipment, 14.5 million refrigerators and 7 million washing machines. With an average product life of 9-10 years, the Delhi-based research and advocacy non-profit Centre for Science and Environment (CSE) estimates that, by 2030, India will generate 14 million tonnes of e-waste, or four times the current volume. The explosion in e-waste has seen a formalisation of the recycling industry, with over 472 units in the country currently. But they are estimated to be handling a mere 15 per cent of the total e-waste generated. It's the informal chain, starting with the friendly neighbourhood kabadiwala, that processes the rest, taking what they need and discarding the rest, posing a hazard for the environment (see *Where does E-waste Go?*). That is the threat that the country needs to address.